HISA

Service Availability & Webhooks

Overview

This guide helps partners integrate with the service availability monitoring and webhook notification system. Partners can:

- **Query service availability** in real-time to display on their platform
- **Receive automatic notifications** when services go down or are restored
- **Proactively manage** service availability on their frontend

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Authentication

All API requests require authentication using a Bearer token in the Authorization header:

```
Authorization: Bearer YOUR_API_KEY
```

Your api keys are shared during onboarding

HOST URL

The service host url is provided during onboarding.

Service Availability API

1. Get All Services Availability

Endpoint: `GET /availability/all`

Request:

```
curl -X GET "{HOST}/availability/all" \
  -H "Authorization: Bearer YOUR_API_KEY"
```

Response:


```
{
  "count": 23,
  "data": [
    {
      "service": "airtime",
      "provider": "mtn",
      "percent": 100
    },
    {
      "service": "airtime",
      "provider": "airtel",
      "percent": 100
    },
    {
      "service": "airtime",
      "provider": "glo",
      "percent": 95
    },
    {
      "service": "data",
      "provider": "mtn",
      "percent": 100
    },
    {
      "service": "data",
      "provider": "airtel",
      "percent": 100
    },
    {
      "service": "data",
      "provider": "glo",
```

```
      "percent": 100
    },
    {
      "service": "cable",
      "provider": "dstv",
      "percent": 100
    },
    {
      "service": "cable",
      "provider": "gotv",
      "percent": 100
    },
    {
      "service": "cable",
      "provider": "startimes",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "aedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "ekedc",
      "percent": 85
    },
    {
      "service": "electricity",
      "provider": "ikedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "ibedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "eedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "phed",
      "percent": 100
    },
    {
      "service": "electricity",
```

```
      "provider": "jedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "kaedco",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "kedco",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "yedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "bedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "yedco",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "aba",
      "percent": 100
    }
  ]
}
```


2. Get Availability by Service Type

Endpoint: `GET /availability/{service}`

Available Services:

- `electricity`
- `airtime`
- `data`
- `cable`

Example - Get All Electricity Providers:

```
curl -X GET "{HOST}/availability/electricity" \
  -H "Authorization: Bearer YOUR_API_KEY"
```

Response:


```
{
  "data": [
    {
      "service": "electricity",
      "provider": "aedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "ekedc",
      "percent": 85
    },
    {
      "service": "electricity",
      "provider": "ikedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "ibedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "eedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "phed",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "jedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "kaedco",
      "percent": 100
```

```
    },
    {
      "service": "electricity",
      "provider": "kedco",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "yedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "bedc",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "yedco",
      "percent": 100
    },
    {
      "service": "electricity",
      "provider": "aba",
      "percent": 100
    }
  ]
}
```

3. Get Specific Provider Availability

Endpoint: `GET /availability/{service}?provider={provider_name}`

Example - Check AEDC Electricity:

```
curl -X GET "{HOST}/availability/electricity?provider=aedc" \
  -H "Authorization: Bearer YOUR_API_KEY"
```

Response:

```
{
  "data": {
    "service": "electricity",
    "provider": "aedc",
    "percent": 85
```

```
  }
}
```

4. Availability Interpretation

Percent	Status	Recommendation
100%	Fully Available	✅ Display service normally
80-99%	Degraded	⚠️ Show warning, allow transactions
50-79%	Limited	⚠️ Show warning, consider disabling
0-49%	Down	❌ Disable service

Best Practice: Disable services below 80% availability to prevent failed transactions.

Webhook Integration

Step 1: Register Your Webhook

Endpoint: `POST /webhooks`

Request:

```
curl -X POST "{HOST}/webhooks" \
  -H "Authorization: Bearer YOUR_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "webhook_url": "https://partner.example.com/webhooks/service-status",
    "auth_token": "your_secure_webhook_secret_token_12345",
    "description": "Production webhook for service status notifications"
```

```
  }'
```

Response:

```
{
  "data": {
    "id": 1,
    "user_id": 123,
    "webhook_url": "https://partner.example.com/webhooks/service-status",
    "is_active": true,
    "description": "Production webhook for service status notifications",
    "created_at": "2026-02-04T15:30:00Z",
    "updated_at": "2026-02-04T15:30:00Z"
  }
}
```

Important Notes:

- `auth_token`: This is YOUR secret token that will be sent in the `Authorization` header when we call your webhook
- `webhook_url`: Must be HTTPS in production
- Keep your `auth_token` secure - it authenticates our requests to your endpoint

Step 2: Implement Webhook Endpoint

Your webhook endpoint must:

1. Accept POST requests
2. Verify the Authorization header matches your `auth_token`
3. Process the webhook payload
4. Return HTTP 200 for success

Step 3: Webhook Payload Structure

Complete Webhook Payload:

```
{
  "event_type": "service_down",
  "service": "electricity",
  "provider": "aedc",
  "previous_state": 100,
  "new_state": 45,
```

```
  "timestamp": "2026-02-04T15:43:27Z",
  "note": "Upstream provider experiencing connectivity issues"
}
```

Field Descriptions:

Field	Type	Description
`event_type`	string	Either `"service_down"` or `"service_restored"`
`service`	string	Service type: `"electricity"`, `"airtime"`, `"data"`, `"cable"`
`provider`	string	Provider name (lowercase): `"aedc"`, `"mtn"`, `"dstv"`, etc.
`previous_state`	integer	Previous availability percentage (0-100)
`new_state`	integer	New availability percentage (0-100)
`timestamp`	string	ISO 8601 timestamp of the event
`note`	string	Optional context about the status change

Event Types:

1. **`service_down`**: Triggered when availability drops from ≥80% to <80%
    - Action: Disable the service on your platform
    - Notify users about the outage
2. **`service_restored`**: Triggered when availability returns from <80% to ≥80%
    - Action: Re-enable the service on your platform
    - Notify users that service is restored

Step 4: Webhook Headers

When we call your webhook, we send these headers:

```
POST /webhooks/service-status HTTP/1.1
Host: partner.example.com
Content-Type: application/json
Authorization: Bearer your_secure_webhook_secret_token_12345
X-Webhook-Event: service_status_change
User-Agent: HISA-Webhook/1.0
```

Header Descriptions:

- `Content-Type`: Always `application/json`
- `Authorization`: Bearer token with your `auth_token`
- `X-Webhook-Event`: Always `service_status_change`

Viewing Webhook Delivery History

Get delivery logs for your webhook:

```
curl -X GET "{HOST}/webhooks/1/deliveries" \
  -H "Authorization: Bearer YOUR_API_KEY"
```

Response:


```
{
  "data": [
    {
      "id": 45,
      "partner_webhook_id": 1,
      "event_id": 23,
      "status": "success",
      "response_code": 200,
      "response_body": "{\"message\":\"Webhook received successfully\"}",
      "error_message": "",
      "attempt_count": 1,
      "created_at": "2026-02-04T15:43:30Z",
      "event": {
        "service": "electricity",
        "provider": "aedc",
```

```
        "event_type": "service_down",
        "previous_state": 100,
        "new_state": 45
      }
    },
    {
      "id": 44,
      "partner_webhook_id": 1,
      "event_id": 22,
      "status": "failed",
      "response_code": 0,
      "response_body": "",
      "error_message": "Request failed: dial tcp: lookup partner.example.com:
no such host",
      "attempt_count": 3,
      "created_at": "2026-02-04T14:20:15Z",
      "event": {
        "service": "data",
        "provider": "mtn",
        "event_type": "service_restored",
        "previous_state": 70,
        "new_state": 100
      }
    }
  ]
}
```


Common Issues & Solutions

Issue 1: Webhook not receiving requests

- ✅ Verify your webhook URL is publicly accessible
- ✅ Check firewall rules allow incoming HTTPS traffic
- ✅ Ensure your server is running and listening on the correct port
- ✅ Verify webhook is active: `GET /webhooks/1`

Issue 2: Receiving 401 Unauthorized

- ✅ Check your webhook endpoint is verifying the Authorization header correctly
- ✅ Ensure the token matches exactly (including "Bearer " prefix)
- ✅ Token is case-sensitive

Issue 3: Webhook timing out

- ✅ Your endpoint must respond within 2 seconds

- ✅ Process webhook asynchronously if it takes longer
- ✅ Return 200 immediately, then process in background

Issue 4: Missing webhook notifications

- ✅ Check delivery history for errors
- ✅ Verify webhook is active (`is_active: true`)
- ✅ Failed deliveries are retried up to 3 times